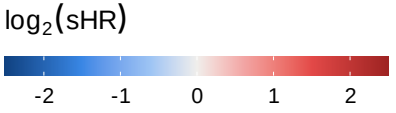
B1 Subdistribution hazard ratios (Fine–Gray) by progression pattern (reference: Sys-TG)

Fine–Gray via survival::finegray with stratum-specific censoring weights; robust SE

	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.05 (0.63–1.77)	1.07 (0.30–3.74)	0.38 (0.21–0.70)**	NE (sparse reference)
IH-NL	0.97 (0.57–1.65)	0.71 (0.19–2.65)	0.33 (0.17–0.62)***	NE (sparse reference)
IH-TG+NL	1.45 (0.82–2.58)	1.29 (0.32–5.19)	0.37 (0.17–0.79)*	NE (sparse reference)
EH-TG	1.06 (0.59–1.93)	2.37 (0.64–8.77)	0.45 (0.23–0.92)*	NE (sparse reference)
EH-NL	0.69 (0.40–1.19)	1.98 (0.58–6.83)	0.63 (0.34–1.14)	NE (sparse reference)
EH-TG+NL	0.79 (0.35–1.78)	1.84 (0.36–9.31)	0.64 (0.25–1.60)	NE (sparse reference)
Sys-TG	Reference	Reference	Reference	Reference (1 event)
Sys-NL	0.72 (0.39–1.33)	3.10 (0.81–11.89)	0.88 (0.45–1.72)	NE (sparse reference)
Sys-TG+NL	0.96 (0.55–1.67)	1.48 (0.41–5.35)	0.90 (0.49–1.66)	NE (sparse reference)
Overall Wald P (df=8)	**P = 0.004**	**P = 0.003**	**P < 0.001**	**P = 0.265**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group. NE: <3 events in the cell or in the reference cell. Overall Wald test (df = 8, robust variance) is reference-independent.